



ya diggg: Master Development Roadmap & Feature Checklist

This document consolidates all strategic decisions, technical specs, and design DNA into a high-level execution plan for ya diggg.

1. Product & Design DNA (The "Soul")

- ☒ **Aesthetic:** 90s Hip-Hop Warmth / Dungeon Family soul [Wikipedia](#) [Bet](#).
- ☒ **Form Factor:** Discreet "Stealth Soul" pocket device.
- ☒ **Material:** Translucent Amber/Orange Polycarbonate (Smoked/Matte).
- ☒ **Interface:** "Disappearing" 1-bit Monochrome Typography.
- ☐ **Tactile Signature:** Define specific Newton (N) force for the orange scan button.
- ☐ **Haptic Signature:** Design distinct vibration "pulses" for (1) Scan Success, (2) Error/Low Light, (3) Scroll "Tick."

2. Hardware & Sourcing Checklist

- ☒ **Display:** 2.7" Sharp Memory LCD (LS027B7DH01) [Digikey](#).
- ☒ **Camera:** 13MP MIPI AF Module (AR1335) [E-Consystems](#)
- ☒ **Processor:** i.MX 8M Nano / PSoC Edge for on-device OCR [Instagram](#).
- ☒ **Battery:** 1500mAh Li-Polymer (12-24h active life) [Tiktok](#).
- ☐ **Thermal Design:** CNC Aluminum internal frame to act as a heat sink for the SoC.
- ☐ **Optics:** Finalize "Stealth" pinhole aperture geometry (Z-height validation).
- ☐ **Sourcing:** Request formal RFQs from **PCBSync** and **CNC Rapid** [Pcbsync](#) [Cncrapid](#)

3. UI/UX & Typography

- ☒ **Font:** 'Crate-Sans' (Swiss geometric, 2px minimum stroke).
- ☒ **Layout:** 'Credits-First' (Personnel, Producer, Sample Lineage).
- ☒ **User Journey:** 3-step "Aim, Scan, Reveal" Quick-Start.
- ☐ **Navigation Logic:** Scroll-dial acceleration curves for long credit lists.
- ☐ **Empty States:** Design UI for "No Match Found" and "Low Light - Retrying"

4. Intelligence & Firmware (The "Brain")

- ☐ **OCR Engine:** Port a lightweight Tesseract custom ML model to the i.MX 8M Nano.
- ☐ **Offline Database:** Design a high-speed, indexed SQLite or NoSQL database for local metadata.
- ☐ **Sync Protocol:** USB-C Mass Storage Class (MSC) or MTP for user library updates.
- ☐ **Power Management:** Deep-sleep state logic (instant-on in <200ms).

5. User Stories & Validation

- ☒ **Story A: The Sample Hunter** (Marcus) - Fast, dirty OCR in warehouse crates .
 - ☒ **Story B: The Blind-Buy Specialist** (Sarah) - High-precision metadata in fair environments
 - ☐ **Story C: The Archivist** - Bulk scanning of collection for cataloging
 - ☐ **Question:** How does the device handle non-Latin alphabet album covers?
 - ☐ **Question:** What is the maximum "Macro" scan distance for legibility?
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6. Sourcing & BOM

- ☒ **Target BOM:** ~\$117.00 .
 - ☒ **Cost Class:** "Playdate-Premium" (\$299-\$399 Retail).
 - ☐ **Final MOQ:** Confirm 500-unit tier costs with CMs.
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Current Status: Transitioning from Design/Research to **Phase 1 Execution (Firmware Architecture & Intelligence Engine)**.